

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (EC)

December 2012

EC 702 Digital Communication & Coding

Date: 30.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the LMS Algorithm [04]
(b) Define Conditional probability with suitable example [03]
- Q - 2 (a) Consider the random process given by $x(t) = A \cos(2\pi f_0 t + \phi)$, where A and f_0 are constant and ϕ is random variable that is uniformly distributed over $[0, 2\pi]$. Calculate the first and second order moments of $x(t)$ using (i) Time average (ii) the PDF of $x(t)$. [07]
- (b) Calculate the minimum Euclidean distance for the adjacent signal for the case of phase modulated signal. [07]

OR

- Q - 2 (a) Explain Continuous-phase FSK [07]
(b) State and prove properties of the Matched Filter [07]
- Q - 3 (a) What is the need of adaptive equalizer? Explain the zero forcing algorithm. [07]
(b) For the duobinary pulses sequence 111010010001101, explain the data detection method for controlled ISI. [07]

OR

- Q - 3 (a) Draw the signal constellation diagram for a 16-QAM and 16 PSK signals with maximum amplitude A . (i) What is the average energy per symbol? (ii) What is the distance between neighboring signal points? [07]
- (b) What is the necessary and sufficient condition for zero ISI for a signal $x(t)$? where $x(t)$ given by [07]

$$x(nT) = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

SECTION – II

- Q - 4 (a) Find the generator polynomial $g(x)$ for a (7,4) cyclic code and find code vector for the following data vector 1111, 0001, 1001, 1000. [04]
- (b) Construct a systematic (7, 4) cyclic code word using generator polynomial $g(x) = x^3 + x^2 + 1$, For data vector $d = 1010$. [03]

- Q - 5 (a) Design the code tree for coder given as $N=3$ & $v=2$ where
 $v_1 = S_1 \oplus S_2 \oplus S_3$, $v_2 = S_1 \oplus S_3$
Take input sequence as 11010 [07]
- (b) Explain binary PSK receiver and QAM receiver with block diagram. [07]

OR

- Q - 5 (a) Explain a discrete-time model for a channel with ISI. [07]
- (b) Explain multicarrier communication system with block diagram [07]
- Q - 6 (a) Derive equation for channel capacity of a Band-Limited AWGN channel [05]
- (b) A zero memory source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1 and 0.08. Find the 4-ary Huffman code. Determine its average word length, the efficiency and the redundancy [03]
- (c) A memoryless source emits messages m_1 and m_2 with probabilities 0.8 and 0.2 respectively. Find the Huffman binary code for this source as well as for its second and third order extensions. Determine the code efficiencies in each case. [06]

OR

- Q - 6 (a) Explain squaring loop with block diagram. [04]
- (b) Explain Linear transversal filter with block diagram. [04]
- (c) Explain multichannel digital communication in AWGN channel [06]